

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 1
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

Electrical Transient Analyzer Program

Short-Circuit Analysis

ANSI Standard

3-Phase Fault Currents

	Swing	V-Control	Load	Total
Number of Buses:	1	0	7	8

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	6	0	0	7

	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Total
Number of Machines:	0	1	0	0	1	2

System Frequency: 50.00
Unit System: Metric
Project Filename: SLD
Output Filename: C:\Users\ASUS\OneDrive\Desktop\Projects\Udemy\Dialog\interior\SLD\test2.SA1S

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 2
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

Adjustments

<u>Tolerance</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Percent</u>
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

<u>Temperature Correction</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Degree C</u>
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Project:	ETAP	Page:	3
Location:	19.0.1C	Date:	09-04-2026
Contract:		SN:	
Engineer:	Study Case: SC	Revision:	Base
Filename:	SLD	Config.:	Normal

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus1	SWNG	11.000	11.000	1	100.00	0.00
Bus3	Load	0.400	0.400	1	100.00	0.00
Bus7	Load	0.400	0.400	1	100.00	0.00
Bus9	Load	0.400	0.400	1	100.00	0.00
Bus10	Load	0.400	0.400	1	100.00	0.00
Bus12	Load	0.400	0.400	1	100.00	0.00
Bus15	Load	0.400	0.400	1	100.00	0.00
Bus16	Load	0.400	0.400	1	100.00	0.00

8 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP .

Project: ETAP
Location: 19.0.1C
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 4
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway		Length							
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable1	1.0MALN3	95	10.0	0	1	75	0.37671	0.08900	0.0000000
Cable3	1.0MALN3	240	30.0	0	1	75	0.15088	0.08710	0.0000000
Cable4	1.0MALN3	16	10.0	0	1	75	2.21892	0.10370	0.0000000
Cable5	1.0MALN3	35	10.0	0	1	75	1.01598	0.09550	0.0000000
Cable6	1.0MALN3	35	12.0	0	1	75	1.01598	0.09550	0.0000000
Cable7	1.0MALN3	240	20.0	0	1	75	0.15088	0.08710	0.0000000

Line / Cable / Busway resistances are listed at the specified temperatures.

Project: ETAP
Location: 19.0.1C
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 5
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

2-Winding Transformer Input Data

Transformer		Rating				Z Variation			% Tap Setting		Adjusted
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z
T1	0.300	11.000	0.400	4.00	10.00	0	0	0	0	0	4.0000

Project: ETAP
Location: 19.0.1C
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 6
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
T1	2W XFMR	Bus1	Bus3	132.67	1326.72	1333.33	
Cable1	Cable	Bus3	Bus7	235.44	55.63	241.92	
Cable3	Cable	Bus3	Bus9	282.90	163.31	326.65	
Cable4	Cable	Bus7	Bus10	1386.83	64.81	1388.34	
Cable5	Cable	Bus7	Bus12	634.99	59.69	637.79	
Cable6	Cable	Bus12	Bus16	761.99	71.63	765.34	
Cable7	Cable	Bus9	Bus15	188.60	108.88	217.77	

Project:ETAP

Location:19.0.1C

Contract:

Engineer:

Filename:SLD

Page:7

Date:09-04-2026

SN:

Revision:Base

Config.:Normal

Study Case: SC

Power Grid Input Data

Power Grid		Connected Bus		Rating		% Impedance 100 MVA Base		
ID		ID		MVAsc	kV	X/R	R	X
U1		Bus1		500.000	11.000	10.00	1.99007	19.90074

Total Connected Power Grids (= 1): 500.000 MVA

Lumped Load Input Data

Lumped Load						Motor Loads							Static Loads	
Lumped Load	Connected Bus	Rating		% Load		Loading		X/R Ratio		% Impedance (Machine Base)			Loading	
		kVA	kV	MTR	STAT	kW	kvar	X"/R	X'/R	R	X"	X'	kW	kvar
Lump3	Bus15	130.0	0.400	40	60	44.2	27.4	2.38	2.38	8.403	20.00	50.00	66.30	41.09

Total Connected Lumped Loads (= 1): 130.0 kVA

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 9
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

SHORT-CIRCUIT REPORT

3-phase fault at bus: **Bus1**

Prefault voltage = 11.000 kV

= 100.00 % of nominal bus kV (11.000 kV)

= 100.00 % of base (11.000 kV)

Contribution		1/2 Cycle					1.5 to 4 Cycle				
From Bus	To Bus	% V	kA	kA	Imag.	kA Symm.	% V	kA	kA	Imag.	kA Symm.
ID	ID	From Bus	Real	Imaginary	/Real	Magnitude	From Bus	Real	Imaginary	/Real	Magnitude
Bus1	Total	0.00	2.616	-26.124	10.0	26.255	0.00	2.612	-26.118	10.0	26.248
Bus3	Bus1	3.07	0.005	-0.011	2.4	0.012	1.34	0.001	-0.005	5.8	0.005
U1	Bus1	100.00	2.611	-26.113	10.0	26.243	100.00	2.611	-26.113	10.0	26.243

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 10
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus3**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus3	Total	0.00	1.196	-10.927	9.1	10.992
Bus7	Bus3	0.00	0.000	0.000	999.9	0.000
Bus9	Bus3	0.77	0.135	-0.315	2.3	0.342
Bus1	Bus3	98.52	1.061	-10.612	10.0	10.665

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 11
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus7**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus7	Total	0.00	2.737	-9.845	3.6	10.218
Bus3	Bus7	17.13	2.737	-9.845	3.6	10.218
Bus10	Bus7	0.00	0.000	0.000	999.9	0.000
Bus12	Bus7	0.00	0.000	0.000	999.9	0.000

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 12
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus9**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus9	Total	0.00	2.590	-9.197	3.6	9.555
Bus3	Bus9	20.85	2.456	-8.880	3.6	9.213
Bus15	Bus9	0.52	0.134	-0.317	2.4	0.345

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 13
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus10**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus10	Total	0.00	4.949	-3.997	0.8	6.361
Bus7	Bus10	61.19	4.949	-3.997	0.8	6.361

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 14
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus12**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus12	Total	0.00	4.803	-6.734	1.4	8.272
Bus7	Bus12	36.55	4.803	-6.734	1.4	8.272
Bus16	Bus12	0.00	0.000	0.000	999.9	0.000

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 15
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus15**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus15	Total	0.00	3.062	-8.139	2.7	8.696
Bus9	Bus15	12.60	2.928	-7.820	2.7	8.350
Lump3	Bus15	100.00	0.134	-0.319	2.4	0.346

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 16
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

3-phase fault at bus: **Bus16**

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base (0.400 kV)

Contribution		1/2 Cycle				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	Imag. /Real	kA Symm. Magnitude
Bus16	Total	0.00	4.764	-4.005	0.8	6.224
Bus12	Bus16	33.00	4.764	-4.005	0.8	6.224

NACD Ratio = 1.00

Indicates a fault current contribution from a three-winding transformer.

* Indicates a fault current through a tie circuit breaker.

If the faulted bus is involved in loops formed by protection devices, the short-circuit contributions through these protective devices will not be reported.

Project:	ETAP	Page:	17
Location:	19.0.1C	Date:	09-04-2026
Contract:		SN:	
Engineer:	Study Case: SC	Revision:	Base
Filename: SLD		Config.:	Normal

Momentary Duty Summary Report

3-Phase Fault Currents: (Prefault Voltage = 100 % of the Bus Nominal Voltage)

Bus		Device		Momentary Duty					Device Capability		
ID	kV	ID	Type	Symm. kA rms	X/R Ratio	M.F.	Asymm. kA rms	Asymm. kA Peak	Symm. kA rms	Asymm. kA rms	Asymm. kA Peak
Bus1	11.000	Bus1	Bus	26.255	10.0	1.438	37.744	64.246			
Bus3	0.400	Bus3	Bus	10.992	9.7	1.432	15.736	26.806			
Bus7	0.400	Bus7	Bus	10.218	3.7	1.167	11.926	20.601			
Bus9	0.400	Bus9	Bus	9.555	3.6	1.159	11.077	19.116			
Bus10	0.400	Bus10	Bus	6.361	0.8	1.000	6.364	9.182			
Bus12	0.400	Bus12	Bus	8.272	1.4	1.012	8.367	12.960			
Bus15	0.400	Bus15	Bus	8.696	2.7	1.090	9.479	16.071			
Bus16	0.400	Bus16	Bus	6.224	0.8	1.001	6.227	9.013			

- Method: IEEE - X/R is calculated from separate R & X networks.

- Generator protective device duty is calculated based on maximum through fault current. Other protective device duty is calculated based on total fault current.

- The multiplication factors for high voltage circuit-breaker and high voltage bus momentary duty (asymmetrical and crest values) are calculated based on system X/R.

* Indicates a device with momentary duty exceeding the device capability.

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: SC

Page: 18
Date: 09-04-2026
SN:
Revision: Base
Config.: Normal

Interrupting Duty Summary Report

3-Phase Fault Currents: (Prefault Voltage = 100 % of the Bus Nominal Voltage)

Bus		Device			Interrupting Duty				Device Capability			
ID	kV	ID	Type	CPT (Cy)	Symm. kA rms	X/R Ratio	M.F.	Adj. Sym. kA rms	kV	Test PF	Rated Int.	Adjusted Int.
Bus1	11.000				26.248	10.0						
Bus3	0.400				10.992	9.7						
Bus7	0.400				10.218	3.7						
Bus9	0.400				9.555	3.6						
Bus10	0.400				6.361	0.8						
Bus12	0.400				8.272	1.4						
Bus15	0.400				8.696	2.7						
Bus16	0.400				6.224	0.8						

- Method: IEEE - X/R is calculated from separate R & X networks.
- HV CB interrupting capability is adjusted based on bus nominal voltage.
- Short-circuit multiplying factor for LV Molded Case and Insulated Case Circuit Breakers is calculated based on asymmetrical current.
- Generator protective device duty is calculated based on maximum through fault current. Other protective device duty is calculated based on total fault current.

- * Indicates a device with interrupting duty exceeding the device capability
- ** Indicates that the circuit breaker has been flagged as a generator circuit breaker. However, ETAP could not detect a single path, without a transformer, to the specified generator. Therefore, this circuit breaker is treated as a regular circuit breaker in short-circuit calculations.
- + The prefault voltage exceeds the rated maximum kV limit of the circuit breaker - The rated interrupting kA must be derated.